

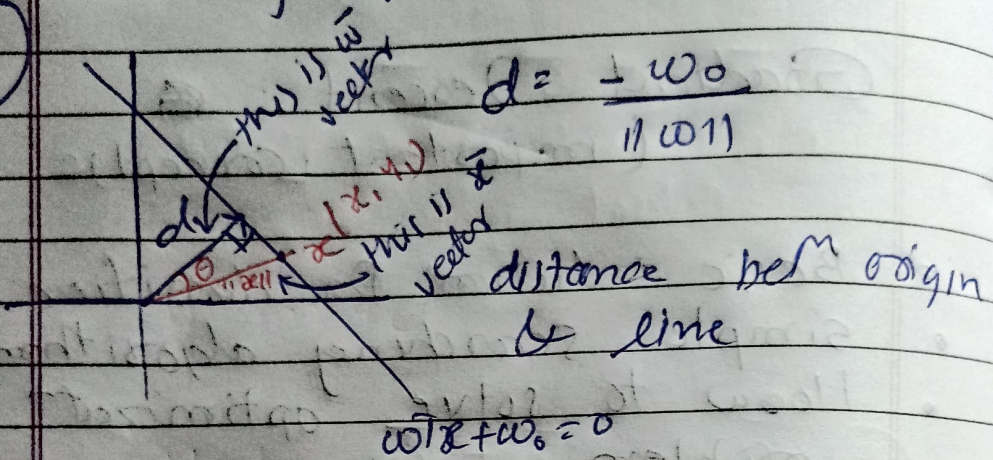
Optimization 1: The Need for Calculus in ML

Gradient Descent (important concept)

- Basic intuition for classifier
- simple searching algorithm
- How to solve optimized problem
- Defining classification problem mathematically
- Relationship betⁿ distance & gain functⁿ
- functⁿ's limits
- continuity
- Domain & range
- Some important functⁿ's

Remaining proof

①



considering w & x are 2 vectors

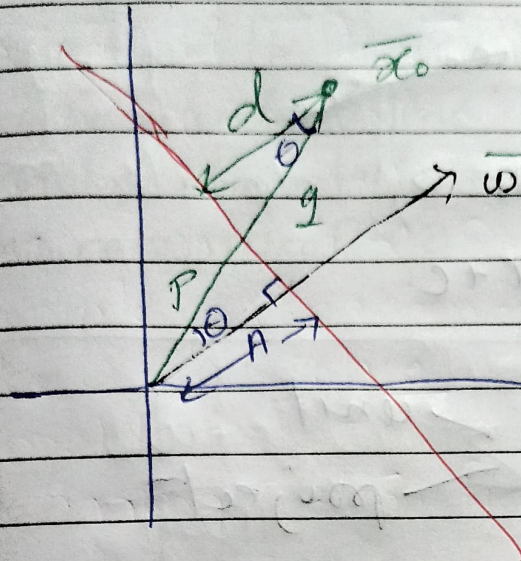
$$\cos \theta = \frac{d}{\|x\|} \quad \cos \theta = \frac{w^T x}{\|w\| \|x\|}$$

from line eqⁿ $w^T x = -w_0$

$$\frac{d}{\|x\|} = \frac{-w_0}{\|w\| \|x\|}$$

$$d = \frac{-w_0}{\|w\|}$$

② Distance of point from line



$$\vec{x}_0 = p + q$$

$$w^T x + w_0 = 0$$

$$A = \frac{-w_0}{\|w\|}$$

$$\cos \theta = \frac{A}{p}$$

$$p = \frac{A}{\cos \theta}$$

$$\|\vec{x}_0\| = p + q$$

$$q = \|\vec{x}_0\| - p = \|\vec{x}_0\| - A / \cos \theta$$

$$\cos \theta = \frac{d}{q} \quad \therefore d = \cos \theta \cdot q$$

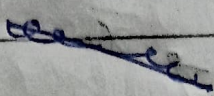
$$d = \left(\|\vec{x}_0\| - \frac{A}{\cos \theta} \right) \cdot \cos \theta$$

$$= \|\vec{x}_0\| \cos \theta - A$$

$$= \|\vec{x}_0\| \frac{w^T \vec{x}_0}{\|w\| \|\vec{x}_0\|} + \frac{w_0}{\|w\|}$$

$$= \frac{w^T \vec{x}_0 + w_0}{\|w\|}$$

Recap

Line 

$$w_1 x + w_0 = 0$$

$$y = mx + c$$

Vectors — unit
 ↓ — project

motion

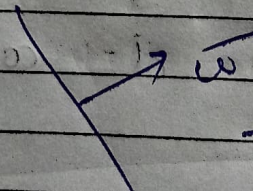
A line

weight $\perp$ line

$$p + q = \|x\| \quad A = q$$

$$p - \|x\| = p - \|x\|$$

$$0.2011 - \|x\|$$

 w
 +ve

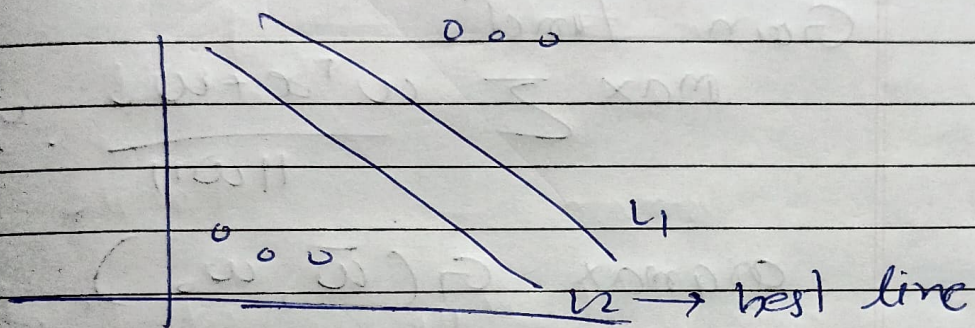
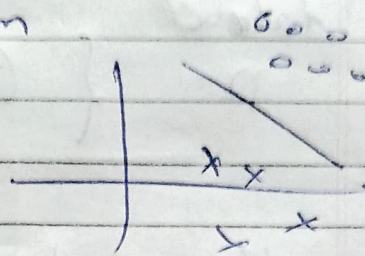
$$-ve \quad w_1 x + w_0 = 0$$

 $w_1 x > 0$ +ve region

 $w_1 x < 0$ -ve region

⑤ perceptron algorithm

- ① collect data
- ② make random line
- ③ Rotate line till
misclassification = 0



perceptron will give you good like line L_2 where misclassification = 0

Now we want L_2 , i.e. optimizing L_1 to get L_2

dataset

$$D = \{ (x_i, y_i) \mid x_i \in \mathbb{R}^d, y_i \in \{1, -1\} \}$$

this is binary classifier.

Gain funcⁿ

$$\max \sum \frac{w^T x + w_0}{||w||} \cdot y_i$$

$$\arg \max G(\bar{w}, w_0)$$

~~argmax~~ example

~~argmax~~ of $-(x-2)^2$ is $x=2$
 as at $x=2$ funcⁿ gives
 max value i.e. 0

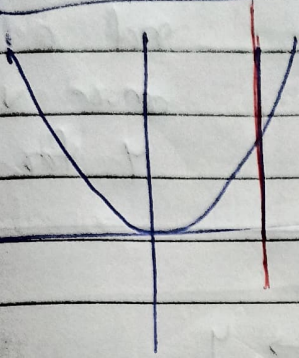
funcⁿ $\rightarrow$ mapping bet^r @ variable
 $f(x) = y$

~~1-1 mapping~~

for every x there
 should be one y
 for every y there

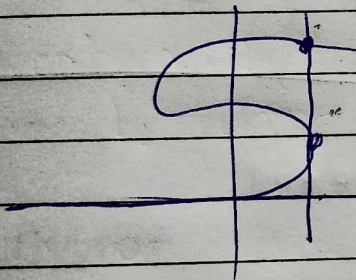
if 2 ~~x~~ values give same y then f
 not funcⁿ might be more x .

how to test
vertical test

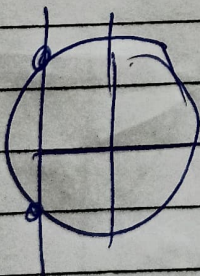


$$f(x) = y = x^2$$

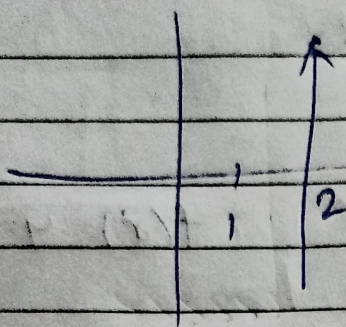
line cuts only 1 point of the curve
if it cuts more the 1 then ~~2~~
no function



not function cutting
2 points



circle $\rightarrow$ not a
function

 $x = 2$

not a function
more value of
 y for $x = 2$

$$x \rightarrow \text{fear} \rightarrow y$$

input

output

Domain

Range

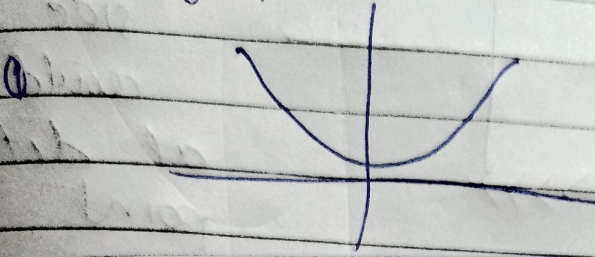


collection of all i/p

collection of all o/p

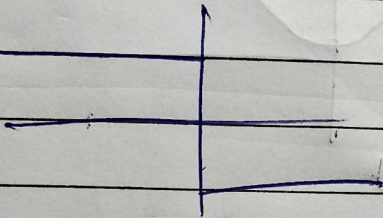
continuous function

draw graph without lifting graph

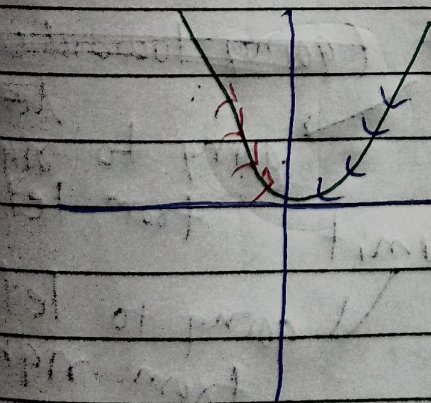


discontinuous function

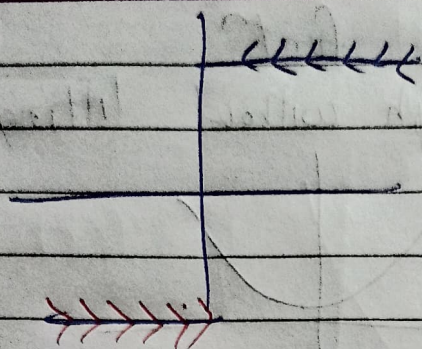
draw graph with lifting graph



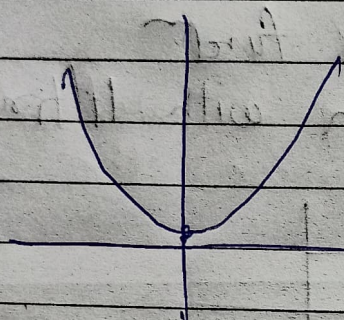
Mathematical way to test continuous & discontinuous function



from both side
one could reach
at some point



from both
side one
could reach
at different
point



$$y = x^2$$

x	y
-1	1
-0.1	0.01
-0.0001	0.000001
$\uparrow$	$\uparrow$
0.1	0.01
0.0001	0.000001

left hand limit ~~going towards 0 from~~
limit

$$x \rightarrow 0^- = 0$$

going to right
from left

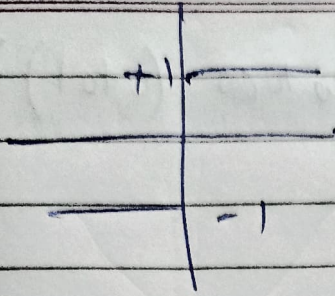
right hand limit

$$x \rightarrow 0^+ = 0$$

going to left
from right

at $x = 0$ $y = 0$

so LHL = RHL
so x^2 is continuous function



$$\lim_{x \rightarrow 0^-} = -1$$

$$\lim_{x \rightarrow 0^+} = 1$$

$$LHL \neq RHL$$

so discontinuous function

find^m Domain range continuous graph

① $y = x$ $-\infty$ to ∞ $-\infty$ to ∞ ✓

② $y = 1/x$ $-\infty$ to ∞ ∞ to ∞ X

③ $y = e^x$ $-\infty$ to ∞ 0 to ∞ ✓

④ $y = |x|$ $-\infty$ to ∞ 0 to ∞ ✓

⑤ $y = \log x$ 0 to ∞ $-\infty$ to ∞ ✓

⑥ $y = \frac{1}{1+e^{-x}}$ (0 to 1) ✓

